stochastic longitudinal mode beating, which results in a spiky temporal intensity profile.

Nanosecond and picosecond pulse shapes were detected by a fast photodiode (ANTEL AR-S1) with a rise time < 100 ps and recorded by means of a 6 GHz oscilloscope (Tektronix DPO 70604). A Gaussian function was fitted to the experimental data to obtain the FWHM time. Femtosecond pulse durations were obtained from a $sech^2$-fit to an autocorrelation trace (APE pulse check).

3.2. Imaging of plasma luminescence and scattering

Open-shutter photography in side view was used to record time-integrated images of the plasma region. The Leica HCX APO L U-V-I 20x, *NA* = 0.5 microscope objective was used for imaging, when the pump pulses were focused through a 40x or 10x objective, and the 40x, *NA* = 0.8 objective was employed, when the pump pulses were focused through the 20x objective. The image produced by the microscope objective and its tube lens was further magnified using an enlarger lens (Nikon, Nikkor 63mm/1:2.8) corrected for 8x magnification. The total magnification was 150 times with 20 x objective and 312 times with 40x objective. Photographs were recorded by a digital camera (Canon EOS 5D, 4368 × 2912 pixels) connected to a PC. In most cases, time-integrated images of single breakdown events were taken but for imaging weak plasma luminescence exposures were integrated over 70-100 laser pulses at ISO 3200. The spectral sensitivity of the chip reaches from 410 nm to 690 nm. All images were recorded using the Auto White Balance function of the camera. Because of the use of water immersion objectives and the large image magnification, the spatial resolution is diffraction limited (0.6 µm for the 20x, *NA* = 0.5 objective and 500 nm wavelength).

On photographs of optical breakdown events produced by 532-nm laser pulses, scattered laser light is visible besides the plasma luminescence. It demarcates the region of primary energy deposition, whereas the plasma luminescence portrays the energy distribution and plasma volume at the end of the laser pulse. A comparison of the extent of both regions provides information on possible energy transport processes during plasma formation. To reveal size and structure of the light scattering region, 532 nm laser radiation was partially blocked by a long-pass colour glass filter (Schott OG 550 with 0.28% transmission at 532 nm) or a dielectric laser mirror acting as a notch filter (Laser Optics 45°, HR at 532 nm, HT around 400 nm and 700 nm, with 0.03% transmission at 532 nm). Scattered laser light could not be detected when UV or IR wavelengths were used, because they are outside the sensitivity range of the Canon EOS 5D camera chip.

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3.3. Determination of breakdown threshold values and threshold sharpness

The onset of bubble formation serves as threshold criterion for optical breakdown. It is detected by a sensitive probe beam scattering technique that also enables to determine the maximum bubble size [117]. For this purpose, a spatially filtered single-frequency cw probe laser beam (CrystaLaser, 660 nm, 40 mW) was adjusted collinear and confocal with the pulsed laser beam. Transmitted probe light was collected by a 10x water immersion objective and imaged onto an AC-coupled amplified photoreceiver (FEMTO, 25 kHz –200 MHz bandwidth), which was protected from the respective pump laser irradiation by appropriate blocking filters.

Breakdown energy thresholds (E_{th}) were determined by counting how frequently bubble formation occurred as the energy was increased from sub-threshold to super-threshold values. Data were binned into small energy intervals (n ≥ 15) with > 20 events per interval, and fitted using the Gaussian error function. E_{th} corresponds to 50 % breakdown probability, and the sharpness S of the breakdown threshold is defined as $S = E_{th}/\Delta E_L$, where ΔE_L is the energy difference between 10 % and 90 % breakdown probability.

The threshold irradiance I_{th} was calculated using the equation

$$I_{th} = \frac{E_{th}}{\tau_L \, \pi \, (M^2 d / 2)^2} \times 3.73. \tag{5}$$

Here τ_L denotes the laser pulse duration, M^2 is the beam quality parameter as listed in Table 1, and d is the diffraction-limited diameter of the Airy pattern arising from focusing a beam with top-hat profile of wavelength λ_L, which is given by $d = 1.22\ \lambda_L/NA$. The factor 3.73 relates the average irradiance values within the pulse duration and focal spot diameter to the respective peak values, which determine the onset of optical breakdown phenomena.

3.4. Determination of bubble size and energy

For large bubbles, the maximum bubble radius, R_{max}, is related to the period of the first bubble oscillation, T_{osc}, by the Rayleigh equation [234]

$$R_{\max} = \frac{T_{\text{osc}}}{1.83} \sqrt{\frac{p_{\text{stat}} - p_{\text{v}}}{\rho_{\text{liq}}}}\ , \tag{6}$$

where p_{stat} = 100 kPa is the hydrostatic pressure, ρ_{liq} = 998 kg/m^3 the liquid density, and p_v the equilibrium vapor pressure inside the bubble (2330 Pa at 20° C). However, for bubble sizes below a few micrometers Eq. (6) must be corrected for the influence of surface tension and viscosity. Surface tension produces a

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pressure scaling inversely proportional to the bubble radius that adds to the hydrostatic pressure, and viscosity also becomes ever more important with decreasing bubble size [63]. Therefore, we determined R_{max} using the Gilmore model of cavitation bubble dynamics, which considers both surface tension and viscosity [42,63] and compared the outcome with the result of Eq. (6). The R_{max} value obtained with the Gilmore model is larger than the value predicted using Eq. (6) by a factor $f = \left[R_{\max,\mathrm{Gilmore}}(T_{\mathrm{osc}}) / R_{\max,\mathrm{Rayleigh}}(T_{\mathrm{osc}})\right]$. For convenience, we provide a fitted function for the dependence of this ratio on T_{osc} that facilitates the R_{max} determination from measured oscillation times:

$$f(T_{osc}) = 0.2697 \times e^{\frac{-T_{osc}}{658.31\mu s}} + 0.5163 \times e^{\frac{-T_{osc}}{140.98\mu s}} + 0.6651 \times e^{\frac{-T_{osc}}{35.071\mu s}} + 0.0709 \times e^{\frac{-T_{osc}}{5410.1\mu s}} + 1. \quad (7)$$

When the influence of surface tension and viscosity on $R_{max}(T_{osc})$ is considered, the light scattering technique provides reliable results even for bubbles with less than 10 ns oscillation time and 100 nm radius. T_{osc} could be measured with better than ± 1 ns accuracy, corresponding to ± 10 nm for R_{max}.

The bubble energy (E_B) is given by

$$E_B = \frac{4}{3}\pi R_{\max}^3 \left(p_{stat} - p_v + \frac{2\sigma}{R_{\max}} \right), \quad (8)$$

where σ = 0.073 N/m is the surface tension of water at room temperature.

3.5. Determination of absorbed energy and its partitioning

The absorbed laser energy was obtained from measurements of the plasma transmittance T_{tra}

$$E_{abs} = E_L (1 - T_{tra}). \quad (9)$$

This approach works well for dense plasmas, where reflection and backscattering are small [218,223,235]. However, close to the bubble threshold, scattering may be more important than absorption, and Eq. (9) should not be used.

For plasma transmission measurements, a 63x water immersion objective (NA = 0.9) was used to collect all transmitted and some forward-scattered laser light onto a calibrated energy meter. Calibration accounted for light losses by reflections at optical surfaces and by absorption in the microscope objective and in water.

For luminescent plasmas, we investigated the partitioning of E_{abs} into fractions going into vaporization of the liquid within the plasma volume, shock wave emission, and bubble formation. The plasma volume V_{plasma} was evaluated from the photographs in side-view, and the vaporization energy calculated as

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